



Research paper

Dealing with zero-inflated data: Achieving state-of-the-art with a two-fold machine learning approach

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ABSTRACT

In many cases, a machine learning model must learn to correctly predict a few data points with particular values of interest in a broader range of data where many target values are zero. Zero-inflated data can be found in diverse scenarios, such as lumpy and intermittent demands, power consumption for home appliances being turned on and off, impurities measurement in distillation processes, and even airport shuttle demand prediction. The presence of zeroes affects the models' learning and may result in poor performance. Furthermore, zeroes also distort the metrics used to compute the model's prediction quality. This paper showcases two real-world use cases (home appliances classification and airport shuttle demand prediction) where a hierarchical model applied in the context of zero-inflated data leads to considerable performance improvements. In particular, for home appliances classification, the weighted average of Precision, Recall, F1, and Area Under the Receiver Operating Characteristic Curve (AUC ROC) was increased by 39%, 49%, 88%, and 48%, respectively. Furthermore, it is estimated that the proposed approach is also four times more energy efficient than the state-of-the-art (SOTA) approach against which it was compared to. Two-fold modeling approaches significantly outperform regular regression, especially when predicting the occurrence of demand events. SOTA results were achieved using Gradient Boosting trees to determine whether an event will occur and Visual Geometry Group (VGG) or Support Vector Regressor (SVR) models for the subsequent classification/regression. The code has been released at two separate repositories.

1. Introduction

Supervised machine learning models exploit relationships between a dependent variable and a set of independent variables encoded in a given dataset. Such models can help explore and understand relationships in data and reduce such relationships to a few parameters. Nevertheless, datasets with abundant zeros in the dependent variable (zero-inflated datasets) are common in certain domains. The abundance of zeroes affects the learning process of machine learning models and renders them incapable of properly predicting zero and non-zero values (Abraham and Tan, 2009). Furthermore, the abundance of zeroes distorts the metrics, making it difficult to assess the quality of the model's predictions (Rožanec et al., 2022).

Multiple approaches have been tried to deal with zero-inflated regression data. Such approaches have been classified into zero-truncated

models (Cohen, 1954) (assume the data comes from a single underlying distribution, where negative values are censored and stacked on zero; e.g., the Tobit model (Tobin, 1958)) and hurdle models (Cragg, 1971; Mullahy, 1986) (two-component mixture models, where one component is concerned with modeling zeroes, while the second one models non-zero occurrences). Nevertheless, Lambert (1992) is recognized among pioneering authors regarding hurdle models, being among the first ones who provided a thorough analysis of zero-inflated models in the presence of covariates and proposed a zero-inflated Poisson regression model to predict defects in a manufacturing setting by modeling a perfect state (where no faults are expected) and an imperfect state (where faults are expected). Since then, many works proposed enhanced approaches. While such an approach has been used and developed in statistics, its application has been scarce when considering machine learning (Abraham and Tan, 2009; Rožanec et al., 2022,

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2023). Furthermore, failure to frame zero-inflated regression problems as a two-fold approach has also affected performance measurement for such models. Finally, while hierarchical approaches have been documented in the literature to address multiple classification problems related to sensor data (e.g., to classify human locomotion (Narayan et al., 2021)), we have not found household classification problems framed this way.

Given the two-fold approach in hurdle models, Abraham and Tan (2009) measured performance considering the accuracy and F-measure to determine classification performance (zero vs. non-zero occurrences) and RMSE to measure the regression performance (including zeroes). Nevertheless, the accuracy metric is inadequate for classification performance in imbalanced datasets. Furthermore, the F-measure is constrained to an arbitrary cut-off threshold. To address this issue, Rožanec et al. (2022) proposed measuring classifier performance considering the Area Under the Receiver Operating Characteristic Curve (AUC ROC, see Bradley (1997)). When measuring the quality of the regression models, RMSE has the desired property of penalizing large errors (e.g., when a non-zero value should have been predicted but was not). Nevertheless, Rožanec et al. (2022) proposed measuring regressors' performance with two variants of the MASE metric (Hyndman et al., 2006). Computing the MASE metrics only on a subset of the test data guarantees that regression performance is measured regardless of zero-inflated data and avoids measurement distortions. Furthermore, the regressor's quality assessment benefits from other MASE metric properties, such as scale invariance, predictable behavior when predicted values are close or equal to zero, symmetry, interpretability, and asymptotic normality. When performing classification in a zero-inflated context, we consider the quality of the classifier is assessed twice with the AUC ROC metric: (i) to measure the first classifiers' capability to predict whether a zero or non-zero value is expected and (ii) to measure the second classifiers' capability to predict a particular class of interest.

The paper describes two real-world use cases where machine learning models are applied to zero-inflated datasets and where these outperform existing SOTA regression models. While the models are different in each case (features are domain-specific, and different models are of interest for each use case), the same approach is used in both cases. In particular, we consider two-fold regression models when predicting airport shuttle passengers' demand and two-fold classifiers to classify home appliances based on their energy consumption. We introduce the two-fold approach in detail in Section 2.

This paper is organized as follows. Section 2 introduces the two-fold approach we followed in this research to achieve excellent results on forecasts for two different real-world use cases with zero-inflated data. In Section 3, we describe the use cases where the abovementioned approach is applied. To the extent of our knowledge, two-fold machine learning models in zero-inflated datasets have not yet been applied to the domains represented by the use cases we tackled. Section 5 introduces the experiments and details the datasets, features, machine learning models, metrics, and hardware used in each case. Section 6 describes and briefly discusses the results we obtained. Finally, in Section 7, we present the conclusions and outline future work.

2. Two-fold forecasting approach

Zero-inflated data is common in specific domains. The abundance of zeroes in the dependent variable is likely to affect the learning process of machine learning models and lead to poor performance. Two-fold approaches are grounded in an observation and two intuitions. We observe that two distinct states can be identified in the dataset: whether the predicted value is zero or different from zero. Furthermore, each state will likely have its own dynamics (e.g., why is no appliance turned on overnight? Why are certain appliances turned on in the morning? How are consumption patterns different between appliances?). Multiple intuitions are associated with these observations: (a) there is probably some domain knowledge that could help us discriminate or predict the

permanence or transition between both states, (b) considering only the time segments associated with the non-zero values could be helpful to produce more accurate predictions for such state outcomes, (c) the overall model performance results from the combined performance on these predicting (i) zero vs. non-zero occurrence and (ii) the value for non-zero occurrence, and (d) measuring the performance of each of those two tasks can help us understand which part of our predictive model needs to be adjusted to achieve optimal results.

The intuitions described above can be illustrated with an example from demand forecasting in a manufacturing setting. Regarding (a), it could be a sound assumption to consider that the demand dates for particular clients are associated with certain shipment days of the week when transport is scheduled to those clients. Such domain knowledge will likely lead to accurate predictions of whether a product will be required on a particular day of the week. Once we have a prediction of whether the demand will exist, we can provide a separate estimate regarding the amount of product required to be shipped. In line with (b), that estimate will likely be best considering past shipments and complementary information, such as open purchase orders from that particular client. These two observations lead us to (c): the overall demand forecasting performance will depend on accurate date predictions and product quantity estimates. To understand which predicted component hurts the overall performance, separate metrics are required to assess the performance of each model - each of them targeting the specific nature of the prediction and performance concerns.

From the example above, we understand that to solve the demand forecasting problem for zero-inflated data, the best results could be achieved by separating the forecasting problem into two parts: (a) forecasting demand occurrence and (b) forecasting demand quantity - each of them considering a model of their, leveraging the data that best informs that particular learning goal, and by optimizing and evaluating them against that particular target (occurrence and quantity estimation respectively). The final forecast will result from combining both models (see Fig. 1).

Analogously, for the use cases presented in this paper, we consider that in the presence of zero-inflated data, they are best solved considering a two-fold approach. When considering shuttle demand, while it is an inherent regression task, in the presence of zero-inflated data, it is best solved by dividing the problem into two parts: (a) a classification model predicting demand occurrence and (b) a regression model predicting the number of passengers requesting the shuttle service. Analogously, home-appliance classification, in the presence of zero-inflated data, is best solved by dividing the problem into two parts: (a) a classifier that will predict whether the home appliance is turned on or off and (b) a classifier that will predict the appliance type based on the consumption patterns. We illustrate the two-fold models for both use cases in Fig. 2.

We root our claims in the empirical evidence of the two use cases described in this research. Nevertheless, our experience in different domains (e.g., Rožanec et al. (2022, 2023)) has shown similar results: decoupling learning into two phases results in a better performance for zero-inflated datasets.

3. Use cases

3.1. Home appliances classification

Environmental concerns are heightened globally due to the rising energy consumption associated with population growth and technological advancements. Because of that, next-generation smart grids allow bidirectional transmission of power and data, creating an automated power grid (Fang et al., 2012) is seen as one of the enablers of increased efficiency. Part of the enabling technologies is also monitoring active home appliances in smart grids. Appliance classification plays a crucial role in smart grid demand response scenarios, which involve adjusting

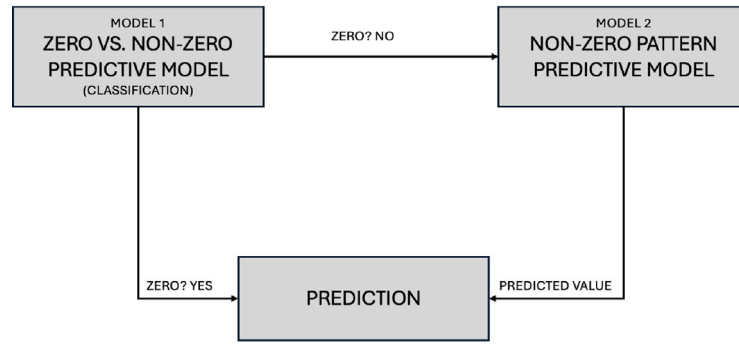


Fig. 1. Schematic diagram of a two-fold forecasting model. First, a prediction is made whether the target value will be zero or different from zero. Depending on that forecast outcome, an additional prediction is made. Each model is optimized for a different target - both meaningful to the final forecast.

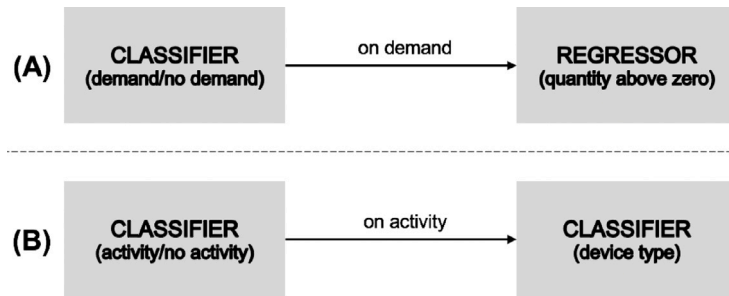


Fig. 2. (A) depicts how a two-fold model is used to predict the airport shuttle passengers' demand: the classifier determines whether some demand will exist, and the regressor predicts the expected number of passengers. (B) depicts how a two-fold model is used to predict the home appliance type based on sensor measurements: the first classifier determines whether some appliance is being used. In contrast, the second one determines the appliance type.

energy consumption based on supply and demand conditions. By providing real-time data on the energy usage of different appliances, the information helps implement demand response strategies by selectively controlling or prioritizing specific devices during peak demand periods or when renewable energy sources are limited. Additionally, accurate appliance classification data can facilitate more precise billing and tariff structures. This enables the implementation of time-of-use tariffs, where different energy prices apply during peak and off-peak hours. Appliance classification helps in accurately attributing energy usage of individual appliances to provide detailed consumption information to consumers and determining fair billing based on each appliance consumption.

Due to the nature of specific home appliances, like toasters, washing machines, and others, a sparse operation cycle can be observed for such devices. For example, a person might use the toaster in the morning to make breakfast, but it would not be used for the rest of the day. Additionally, constant deep learning model inference can result in a higher energy consumption (Cerar et al., 2022), compared to the energy savings enabled by the technology.

This use case demonstrates on a real-world dataset how a simple two-fold machine learning approach can enhance the performance of the classification of home appliances in next-generation smart grids. 75% of the sensor measurements correspond to device inactivity, and therefore, just 25% are left to learn characteristic patterns that enable discriminating between devices. Fig. 3 shows how recurrence plots change when only activity periods are extracted. Fig. 4 shows recurrence plots (whole measurements vs. activity only) for five device types. It can be observed that recurrence plots where only periods of activity are considered are substantially different from each other vs. those that consider the whole measurement period.

3.2. Shuttle transfer demand

Shuttle transfers provide convenient and efficient transportation to individuals or groups of passengers between airports, hotels, and

other designated locations. Passengers are pooled in vehicles based on their departure time, flexibility, and routing preferences, resulting in efficient and cost-effective transportation solutions. To optimize sales and operations, it is crucial to have advanced knowledge of the number of passengers. This information offers two significant advantages: enabling more efficient vehicle allocation and facilitating dynamic pricing strategies. For instance, having information about the expected number of passengers booking a particular route at a specific time allows for precise vehicle allocation based on matching the capacity requirements. In addition, when the number of passengers that will travel together on a particular day and hour is known during the booking process, it enables a fairer distribution of the costs associated with the ride. Unfortunately, long-term prediction of the number of passengers is relatively challenging due to its susceptibility to factors like holidays, plane departures, and arrivals. Furthermore, passengers often book their rides months, weeks, or even just a few days in advance, making it difficult to allocate enough vehicles on time and offer a fair fixed price to early bookers without knowing how many people will be pooled together in a single vehicle.

The paper focuses on a real-world shuttle company for which three specific routes ($A \rightarrow B$, $A \rightarrow C$, and $A \rightarrow D$) were selected. In these routes, the shuttles transport passengers between 15% and 50% of the time (leading to a sparsity between 50% and 85%). Accurately predicting the number of passengers traveling between two cities and the optimal timing is critical to appropriate planning. This involves predicting the occurrence of at least one reservation (i.e., whether a vehicle will be required to travel) and the number of passengers to be transported (i.e., the number of reservations). Due to these two reasons, the problem is a natural fit for a two-fold regression.

4. Related work

This section briefly describes scientific works closely related to the use cases presented in Section 3. In particular, we introduce surveys

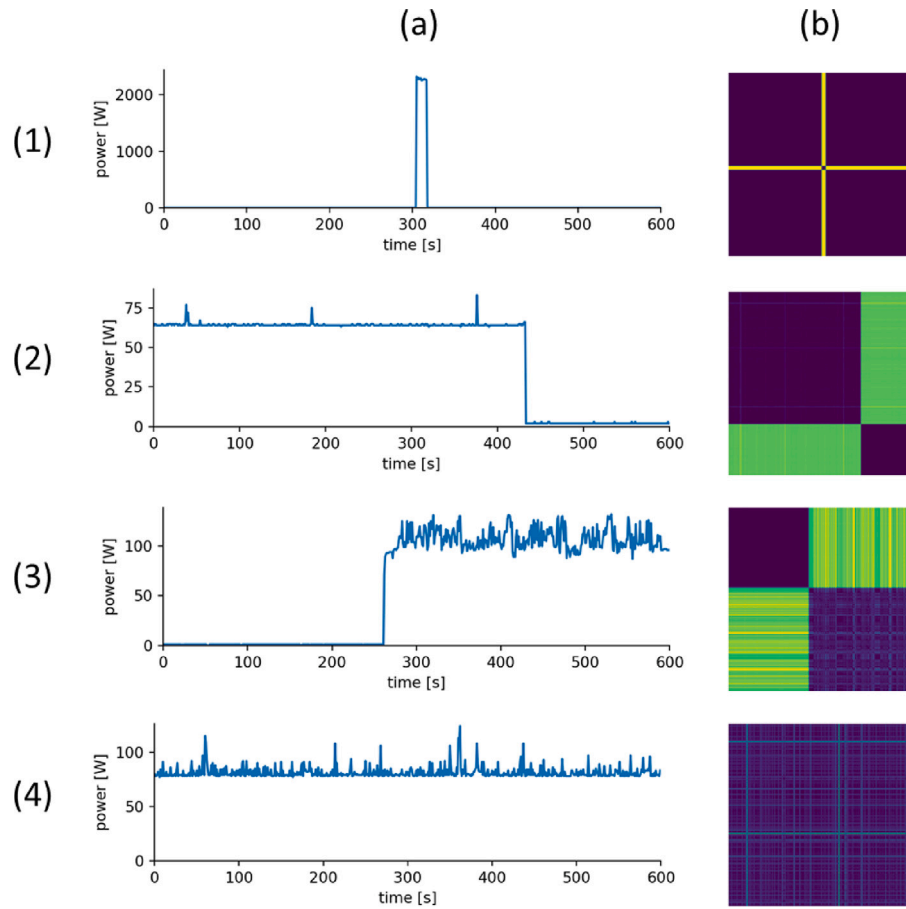


Fig. 3. The plots represent boiler samples from the UKDALE dataset. The samples refer to the (1) HEKA, (2) HTPC, (3) TV, and (4) home server (NAS) appliances. Column (a) shows the time series reporting power consumption, while column (b) displays recurrence plots capturing the time series information. The recurrence plots are used as an input to the VGG11 model.

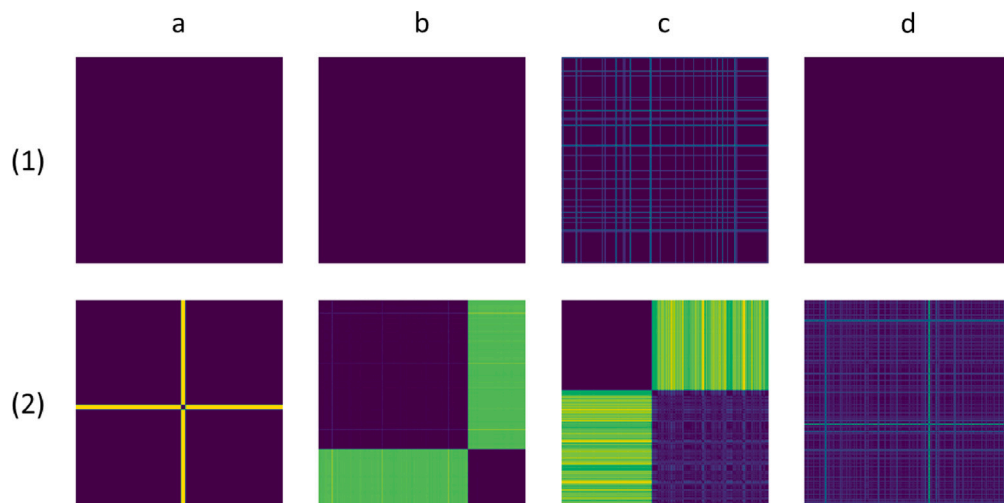


Fig. 4. The recurrence plots represent different appliances from the UKDALE dataset: (a) HEKA, (b) HTPC, (c) TV, and (d) home server (NAS). The recurrence plots at the top take into account the whole measurement period, while those at the bottom were computed considering time series segments related to activity periods only.

associated with the specific domain area the use case relates to, along with other relevant works, and describe the research gaps we identified and that we address through our research.

4.1. Home appliances classification

Time series are defined as an ordered sequence of observations of finite length taken through time, typically taken at equal-spaced time intervals (Abanda et al., 2019; Gupta et al., 2020). Data mining problems related to time series have been considered among the most challenging problems in the past three decades and are among the hardest data mining research topics nowadays (Ismail Fawaz et al., 2019; Foumani et al., 2023). The growing amount of recorded data increased research in this field and originated many novel methods to solve time series-related tasks, among which time series classification has a prominent place (Abanda et al., 2019). Time series classification relevance stems from the fact that any data where there is some notion of order can be cast as a time series classification problem and that time series can be found in domains such as cybersecurity, human activity recognition, and forestry monitoring, among others (Hemmerling et al., 2021; MontazeriShatoori et al., 2020; Suh et al., 2023). Multiple surveys have been written on this topic, providing a comprehensive overview and evaluation of time series classification techniques. Bag-nall et al. (2017) provide a general overview of time series classification techniques that require feature engineering. The authors introduce a taxonomy and classify approaches into six categories (approaches comparing (i) whole series, (ii) intervals, (iii) shapelets (short patterns), (iv) dictionary methods (consider the frequency of repetition of subseries), (v) combinations (combine multiple of the abovementioned approaches), and (vi) model based (fit generative models and measure the similarity between series)). A complementary work was done by Gupta et al. (2020), who focused on approaches for early time series classification. On the other hand, Abanda et al. (2019) provides a comprehensive review of distance-based techniques, categorizing them into three types of approaches: (i) k-NN, (ii) methods that rely on distance features, and (iii) methods that rely on distance kernels. Ismail Fawaz et al. (2019) provide an in-depth review of deep learning techniques for time series classification while also assessing the performance of many of them on a common dataset. This survey is complemented by the work by Foumani et al. (2023), who surveyed more recent deep learning methods for time series classification. Given many open time series datasets do not have massive amounts of samples that would enable using deep learning techniques (Alzubaidi et al., 2023). This void was addressed by Iwana and Uchida (2021), who focused on reviewing data augmentation techniques that could be used for time series classification. When considering time series classification of sensor data for appliance recognition, two distinct cases are frequently considered: non-intrusive and intrusive load monitoring. Among survey papers introducing the topic, Gopinath et al. (2020) provide a comprehensive review of non-intrusive load monitoring for appliance recognition for energy management. The authors distinguish multiple types of appliances, categorize feature types into three groups (steady-state signatures, transient-state signatures, and non-traditional features), survey supervised and unsupervised learning techniques used in this domain, and describe how the performance evaluation of such approaches is done. Less recent works include the surveys by Zoha et al. (2012) and Klemenjak and Goldsbrough (2016), which provide complementary and valuable insights. Works on intrusive load monitoring are scarcer. Nevertheless, a brief overview is provided by Ridi et al. (2014). Among surveys describing both approaches, we find the works by Susto et al. (2018) and Franco et al. (2021). In particular, Susto et al. (2018) describe time series classification approaches for power systems data, proposing a taxonomy of time series classification approaches (feature-based and distance-based) and highlighting their application. From the related work we have studied and referenced above, we have found that none of the works considered time series with zero-inflated data or neglected to consider it as such. This work aims to bridge the gap by providing a novel perspective on sensor data and a different approach to dealing with zero-inflated sensor data.

4.2. Shuttle transfer demand

Accurately predicting transportation demand is key to guaranteeing a good customer experience while optimizing operational costs. Forecasting for highly variable or low demands poses additional challenges and frequently results in poor forecasting efficiency. Many demand-responsive approaches have been developed to address that concern, summarized in a comprehensive survey by Vansteenwegen et al. (2022). The authors consider three levels of responsiveness (dynamic online, dynamic offline, and static) and mostly report heuristic and simulation methods. This survey complements the great work done by Ibarra-Rojas et al. (2015), who provide an overview of the planning, operation, and control of bus transport systems, describing problems and solutions envisioned when considering the planning horizon. They consider the solutions at three levels: strategic, tactical, and operational. Nevertheless, neither author addresses the transport demand specific to passengers traveling from/to airports and almost exclusively focuses on heuristic approaches. Banerjee et al. (2020) focused on passenger demand forecasting in scheduled transportation, and while surveys heuristic and some machine learning techniques, these assume demand events are not scarce. Similarly, Rodrigues et al. (2020) used ARIMA and artificial neural network models to predict taxi demand between particular areas of Lisbon, Portugal, including demand for taxi services heading from/to the airport. Among works related to transportation demand between the city and an airport we also find the research done by Li and Sheng (2016), who studied the mode choice behavior of inter-city passengers behavior between four typical city pairs located in the Beijing-Guangzhou corridor, China; and the research done by Truong and Ngoc (2020), who forecast city-airport transportation demand to guide urban transport planning. While the abovementioned works enrich the understanding of the complex problem of airport shuttle demand forecasting, we found they mostly focus on heuristic approaches. Furthermore, none considered time series with zero-inflated data or neglected to consider it as such. The experiments we conducted and the research results we report aim to bridge this gap by providing a novel perspective on airport shuttle demand forecasting when dealing with zero-inflated data.

5. Experiments

5.1. Home appliances classification

This research contrasts the usual SOTA approach to home appliance classification with a two-fold classification approach. The two-fold machine learning models consist of (a) a classification model to predict home appliance activity and (b) a classification model to discriminate the kind of appliance used. While the first classifier is trained as a binary classifier over all the data (considering two cases: activity or no activity), the second classifier is trained only on time series segments displaying appliance activity.

Dataset: The UK-DALE (Kelly and Knottenbelt, 2015) dataset was used, which has real-world measurements of power consumption from 54 appliances collected over 655 days in five households in the United Kingdom. Measurements were collected with a sampling time of six seconds, ideally resulting in 600 measurements per one-hour time window. All time series samples from the dataset were divided into sub-intervals with a one-hour fixed window length, and samples with missing data were discarded. Appliances that had less than ten percent of samples of the largest class were also discarded. Using domain knowledge, fridges and freezers were merged into a single category (i.e., “fridge/freezer”), and the same was done for toasters and kettles (i.e., “HEKA”), given they display similar patterns. As a result, twelve different appliance classes were considered for the experiments. The final dataset contains 1 275 508 samples, where 25% (319 830) samples contain actual activity.

Features: To identify the activity of home appliances, raw measurements from the preceding hour were used. To determine which

appliance is active, time series data were transformed into images leveraging recurrence plots (Eckmann et al., 1995; Bertalaníč and Fortuna, 2024; Matindife et al., 2021). A deep learning architecture was used to extract features and determine the type of appliance based on such images.

Machine learning models: Two different machine learning models were considered. First, a VGG11 (Simonyan and Zisserman, 2014) architecture was used to discriminate between home appliances based on recurrence plots. VGG11 presents a good choice for tasks where simplicity, flexibility and computation efficiency (within the VGG family) are more important than cutting-edge performance. This combination has already shown SOTA results across several domains (Bertalaníč et al., 2022; Pirnat et al., 2024). The study of idle states of devices has shown that some devices, even inactive, can still consume some power when idle. Given the range of idle power consumption varies between devices, a threshold method cannot correctly detect idle or active states for an extensive range of devices. Therefore, an XGBoost classifier was trained to predict whether an appliance was active during the observation window. The appliance type was determined with a VGG11 model, leveraging a recurrence plot built considering only the appliance activity period.

Model training: The XGBoost classifier was used with default hyperparameters (XGBoost v1.7.6 implementation). The VGG11 architecture was trained from scratch using Adam optimizer with a learning rate of 0.001, batch size 128, and cross-entropy as a loss metric. The training duration was limited to 100 epochs and early-stopping if the evaluation score would not improve for 10 epochs. The classes were weighted during the training procedure to mitigate class imbalance. The models were trained on shuffled data using 20 stratified K-Folds. For each fold, the model was initialized with a different seed.

Metrics: The models' performance was measured with four metrics: F1, precision, recall (the three of them reported in previous works - we report them to provide comparability), and the AUC ROC, which summarizes the overall models' discriminative power while not being affected by the class imbalance. The AUC ROC is reported twice for the two-fold model: once to measure how accurately the model discriminates between activity vs. non-activity periods and once to assess the overall discriminative capability when considering appliance types.

Hardware: All tests were conducted using a computer equipped with two AMD(R) EPYC(R) 75F3 CPUs operating at 3 GHz, 1TB of RAM, 3 g.40gb instance of NVIDIA(R) A100, and running Ubuntu Server operating system.

Model complexity and theoretical energy consumption: According to the guidelines considered in Bertalaníč et al. (2022), the complexity of the model was measured in floating point operations (FLOPs) needed for prediction. Based on the calculated FLOPs and the provided hardware details, the theoretical energy consumption per prediction (TEC) can be calculated. TEC was calculated by Eq. (1), where FLOPs/Watt represents floating point operations per second per watt. Theoretical FLOPs/Watt for an A100 Nvidia GPU double precision (float64) is 38.8 GFLOPs/Watt.

$$TEC = \frac{FLOPs}{FLOPs/Watt} \quad (1)$$

5.2. Shuttle transfer demand

Regression models that achieved SOTA performance in a particular dataset against two-fold machine learning models inspired by zero-inflated regression approaches. The two-fold machine learning models consist of (a) a classification model to predict when zero or non-zero values occur and (b) a regression model to predict the non-zero values. While the classification model is trained as a binary classifier over all the data (considering two possible classes: zero or non-zero), the regressor is trained only on non-zero data to avoid issues derived from zeroes occurrence. The following comparisons are made in all cases: (a) zero-inflated regression models vs. traditional regressors that achieved

SOTA performance on such datasets and (b) local vs. global models for classification and regression. While local models are trained only on the data of a particular destination, global models are trained on the data of the three destinations under consideration.

Dataset: The dataset comprises hourly demand data collected over four years. Due to its relatively low demand density, the data is grouped into intervals of three hours. As a result, when making demand forecasts, the passenger count departing from a particular city to some destination is anticipated within three hours. After consulting with planners, the three-hour aggregation interval was selected as the most practical choice for operational decision-making.

Features: A set of 14 hand-crafted features were created. A feature was created to indicate whether the day corresponded to a holiday. Temporal attributes such as hour, day of the month, month, and day of the week were considered. Cyclical timestamp features were used to capture cyclic temporal patterns, effectively encoding recurring patterns within intervals such as quarter days, half days, full days, and weeks. Furthermore, a variable was used to discriminate between weekdays and weekends. Such features were selected based on existing literature when modeling temporal forecasting problems (Zhou et al., 2019; Balderas-Díaz et al., 2024; Petelin et al., 2023).

All features were scaled using a standard scaler, which subtracts the mean and scales values to unit variance. All the regressors had their targets scaled using the $\log(1 + x)$ transformation before training. After making a prediction, its value was rescaled to its original values using the $e^x - 1$ transformation.

Machine learning models: The experiments considered three regressor machine learning algorithms trained over all of the time series values: A Histogram-Based Gradient Boosting (HGB) tree regressor, a Multi-Layer Perceptron (MLP), and a Linear Regression (LR). The LR was considered due to its simplicity. The two-fold models considered a Histogram-Based Gradient Boosting (HGB) tree classifier combined with a machine learning regressor (either LR, MLP, or Support Vector Regressor (SVR)). While the classifier was trained on the whole time series, the regressors were trained only on non-zero data. The models were selected due to recommendations from literature on good performance in many tabular datasets (Grinsztajn et al., 2022; McElfresh et al., 2024). The Youden index (Youden, 1950) (defined as $J = sensitivity + specificity - 1$) was computed to determine the best threshold cut-off for every classification model. Given the high sparsity levels in the data, a zero predictor was considered a baseline model. The default hyperparameters provided by *scikit-learn* version 1.3.0 were used in the experiments. For the MLP regression task, a model with a single hidden layer comprising 100 neurons, a ReLU activation function, and an Adam optimizer was used. A model with a radial basis function kernel and a regularization set to 1.0 was employed for the SVR analysis. With a histogram-based gradient boosting tree, a learning rate of 0.1 was used with a maximum iteration limit of 100. Log-loss and squared error loss were applied to classification and regression models, respectively.

A time series cross-validation setting (expanding window technique) was considered, forecasting 24 h ahead. The models' performance was tested in the last month contained in the dataset. For each new day considered in the test set, the models were retrained with the information available up to 24 h before the predicted time slot. The experiments were repeated twenty times to ensure the results were reliable and accounted for the stochasticity of some machine learning algorithms.

Metrics: Two metrics were considered for measuring the models' performance: AUC ROC (to measure how well the models predict when a demand event will occur) and MASE (to measure how well the model estimates the demand quantity). Furthermore, Wilcoxon paired rank tests were performed to determine the statistical significance of the results.

While two-fold models provide classifier predictions for assessing the AUC ROC, this is not true for usual regressors. Therefore, three

strategies were devised to turn regression predictions into predictions signaling event occurrence. First, consider any value below one a non-demand event, and every value equal or greater than one to signal a demand event (AUC ROC_{floor}). Second, consider any value equal to or below 0.5 to be a non-demand event and every value above that to be a demand event (AUC ROC_{round}). Finally, consider only zero values as non-demand events and any value above zero as a demand event (AUC ROC_{ceil}).

MASE was computed with the following equation: $MASE = \frac{\frac{1}{J} \sum_j |e_j|}{\frac{1}{J} \sum_j |y_j - \bar{y}|}$

Following Rožanec et al. (2022), two MASE variants were considered (namely MASE_I and MASE_{II}). MASE_I was computed on the test data that results from ignoring zero values. On the other hand, MASE_{II} was computed considering all points where either a non-zero value was observed or the classification model predicted non-zero occurrence. While MASE_I describes the model's performance when true events occur, MASE_{II} provides insights into the cost of false positives. When the model predicting event occurrence performs perfectly, MASE_I equals MASE_{II}.

Hardware: All tests were conducted using a computer equipped with an Intel(R) Xeon(R) CPU E5-2680 v3 operating at 2.50 GHz, 1 TB of RAM, and running the Ubuntu operating system.

6. Results and discussion

6.1. Home appliances classification

Table 1 presents the home appliance classification results. At the two-fold models, the XGBoost classifier predicting whether some home appliance activity is detected achieved an AUC ROC of almost perfect 100%. The home appliance classification outcomes show that using a two-fold approach enhanced the overall model's discriminative performance by AUC ROC 0.3173 points on average and up to 0.3703 AUC ROC points in the best case, leading to almost perfect classification performance in many cases. The rest of the metrics mirrored this improvement. In particular, the weighted average between both approaches showed Precision, Recall, and F1 increased by 0.2316, 0.2707, and 0.384 points, respectively. The cases where Precision increased the most were HEKA (0.3081 points increase) and light (0.2912 points increase). Recall improvements were notorious. E.g., HEKA increased recall from 0.0844 to 0.8287 (0.7443 points), the HTPC Recall increased from 0.1220 to 0.8651 (0.7431 points), and the microwave from 0.0786 to 0.8617 (0.7831 points). Substantial Precision and Recall improvements were also mirrored in F1 scores. For example, the washer-dryer improved the F1 score by 0.6333 points, the desktop computer by 0.5683 points, and the computer monitor by 0.5733 points.

On top of the SOTA results achieved by the two-fold approach, we expect it to show lower energy consumption, given that the deep learning model (more resource intensive than the XGBoost) is utilized only when some appliance activity is predicted. Following the method proposed by Bertalaníč et al. (2022), the model approach requires about 13.3 GFLOPs per prediction. The dataset used consisted of 1 275 508 samples, of which 319 830 were labeled with activity. Without the two-step classification process, TEC would consume ≈ 437.2 kJ of energy for all 1 275 508 samples, compared with ≈ 109.6 kJ for activity classes only. This means that about four times less energy is consumed, making the approach proposed in this paper much more energy efficient, considering that the computational complexity of XGBoost is negligible compared to the deep learning approaches.

6.2. Shuttle transfer demand

The performance for the best and second-best models, along with the zero predictor model and best-performing regressor (that is not a two-fold model), are presented in Table 2. In all cases, the two-fold

regression models led to the best results. In particular, we observed that the best results were achieved by the local two-fold models and followed by the global two-fold models. In most cases, using data from other Source→Destination paths (global models) negatively affected the predictor outcomes. An interesting finding about the regressive power of the models is that most of them, while having predictive power about when a demand event will take place, had little predictive power regarding the number of passengers that will require the service. This is evident from the MASE_I values, usually above one. Therefore, two conclusions can be drawn: (i) usually, better results can be obtained by just predicting the average number of passengers as learned from the train set, and (ii) additional features are required to signal better the number of potential passengers for a given route. It must be noted that the SVR regressor trained on top of the classifier for two-fold models performed best in most cases and that the difference between regressors trained at the two-fold models was statistically significant when tested with the Wilcoxon signed-rank test at a p -value = 0.05 for two routes. Furthermore, the models' performance measured across the different metrics for the models displayed in Table 2 is considered statistically significant when performing the Wilcoxon signed-rank test at a p -value = 0.05.

While the results in Table 2 show the best AUC ROC obtained for each regression model, these could correspond to different class attribution strategies as detailed in Section 5.2. To illustrate this, we provide a few examples in Table 3. The table shows that different class attribution strategies may result in different performance measurements. For example, while AUC ROC_{round} score for the LR on all values model is 0.7079, the same regression predictions led to AUC ROC_{floor} and AUC ROC_{ceil} scores of 0.5791 and 0.5000. While the ceiling of the values removed any intuition about when the demand would occur, rounding them strengthened the signal and led to competitive results.

Regarding computational cost, the two-fold approach is roughly twice as expensive to train, as we are effectively training one classifier and one regressor compared to classical approaches where a single regressor is trained. Additionally, during inference, two models are required to make a prediction, resulting in a slightly larger overhead than regular approaches.

7. Conclusions and future work

This research addressed the problem of zero-inflated data by considering two-fold machine learning models to solve real-world regression and classification problems.

The results show that such an approach achieves excellent results when predicting shuttle transfer demand. While regular regression models sometimes learn when certain shuttle transfers will be required, they may fail to do so, resulting in spurious predictions. On the other hand, two-fold models adequately captured the demand event occurrence in all cases. Nevertheless, when predicting the number of passengers per shuttle, the models lagged compared to predicting the average number of passengers observed in the training period (MASE was slightly above 1 in most cases). Therefore, better features are required to capture the dynamics that influence the number of travelers.

The two-fold approach achieved SOTA results for home appliance classification. By learning activity and non-activity patterns, the data used to classify appliance types is much more informative, leading to better classification results. In particular, when comparing the weighted average of Precision, Recall, F1, and AUC ROC, we observed performance increased by 39%, 49%, 88%, and 48%, respectively. Furthermore, it is estimated that this approach is four times more energy efficient than the SOTA method against which it was compared.

A two-fold approach can be particularly advantageous in cases where the data exhibits a high proportion of zeros and traditional modeling approaches struggle to capture this characteristic effectively. The success of reformulating a problem using a two-fold framework

Table 1

UKDALE classification results. In the last column, AUC uses a one-vs-rest approach and weighted average.

Class	Sparse time series data				Time series data			
	Precision	Recall	F1	AUC ROC	Precision	Recall	F1	AUC ROC
HEKA	0.6583	0.0844	0.1496	0.6230	0.9663	0.8287	0.8922	0.9933
HTPC	0.5817	0.1220	0.2017	0.7788	0.7320	0.8651	0.7930	0.9768
boiler	0.6282	0.2348	0.3418	0.7833	0.8149	0.8058	0.8103	0.9754
computer monitor	0.5510	0.0145	0.0283	0.6350	0.6198	0.5844	0.6016	0.9749
desktop computer	0.7164	0.0282	0.0542	0.6351	0.7366	0.5390	0.6225	0.9739
fridge/freezer	0.8039	0.3981	0.5325	0.8461	0.9441	0.9737	0.9587	0.9975
laptop computer	0.5722	0.0350	0.0659	0.6307	0.7052	0.6905	0.6978	0.9796
light	0.5422	0.9871	0.6999	0.6308	0.8333	0.8437	0.8385	0.9679
microwave	0.5674	0.0786	0.1381	0.6400	0.7513	0.8617	0.8027	0.9933
server computer	0.5985	0.2464	0.3491	0.8407	0.8201	0.7297	0.7723	0.9920
television	0.5538	0.0598	0.1080	0.7848	0.7415	0.5863	0.6548	0.9641
washer dryer	0.8337	0.0248	0.0481	0.6296	0.9337	0.5364	0.6814	0.9646
Weighted avg.	0.5938	0.5526	0.4371	0.6613	0.8254	0.8233	0.8211	0.9786

Table 2

Best and second-best local and global models and the zero predictor model are displayed. The best results are bolded, and the second-best are underlined.

Source→Destination	Type	Model	Best AUC ROC	MASE _I	MASE _{II}
$A \rightarrow B$	–	Zero Predictor	0.5000	0.6875	0.6875
	local	Classifier (HGB) + LR	0.6908	1.3766	1.4139
		Classifier (HGB) + MLP	0.6908	1.3766	1.4139
		Classifier (HGB) + SVR	0.6908	1.3766	1.4139
		HGB Regression on all values	0.5066	0.9338	0.9512
	global	<u>Classifier (HGB) + LR</u>	<u>0.5903</u>	<u>1.3375</u>	<u>1.3375</u>
		HGB Regression on all values	0.5625	1.5574	3.7406
$A \rightarrow C$	–	Zero Predictor	0.5000	0.9120	0.9120
	local	Classifier (HGB) + SVR	0.7882	1.3118	1.4945
		HGB Regression on all values	0.7396	0.9303	0.9391
	global	<u>Classifier (HGB) + SVR</u>	<u>0.7495</u>	<u>0.9469</u>	<u>0.9927</u>
		LR on all values	0.7327	0.8468	0.8833
	–	Zero Predictor	0.5000	0.6883	0.6883
$A \rightarrow D$	local	Classifier (HGB) + SVR	0.6931	1.5852	1.7654
		HGB Regression on all values	0.6736	1.1372	1.1912
	global	<u>Classifier (HGB) + SVR</u>	<u>0.6836</u>	<u>1.0640</u>	<u>1.0976</u>
		HGB Regression on all values	0.6670	1.0634	1.0876

Table 3

An example of AUC ROC values that result from different class attribution strategies.

Source→Destination	Type	Model	AUC ROC _{round}	AUC ROC _{floor}	AUC ROC _{ceil}
$A \rightarrow C$	–	Zero Predictor	0.5000	0.5000	0.5000
	local	LR on all values	0.7079	0.5791	0.5000
		MLP on all values	0.7189	0.5907	0.5000
		HGB Regression on all values	0.7396	0.6785	0.5000

largely depends on factors such as the prevalence of zeros in the data, the underlying structure of the problem, and the choice of machine learning algorithms employed.

The two-fold approach can be applied to other domains and applications that suffer from zero-inflated data. While the features used in each case may be domain-specific, exploring approaches where generic features could be derived would be interesting, enabling domain adaptation and greater generalization. A possible direction of research in this regard would be to explore representing data as graphs and to leverage machine learning on graphs for the two-fold approach. Since graphs can encode very different data while providing a similar representation, they could eventually enable domain adaptation and better generalization across domains.

We also envision multiple research directions associated with the use cases. In particular, future work regarding home appliance classification will focus on complementing the two-fold approach with regression models that leverage graph representations of time series

to reduce the computational cost and obtain more energy-efficient machine learning models while still achieving SOTA results. Regarding shuttle demand prediction, we envision at least four lines of work: (a) dataset enrichment and feature engineering (e.g., considering passengers' most frequent travel destinations) to enhance the existing models, (b) building a graph with geographic and temporal data regarding the shuttle routes and apply graph machine learning algorithms to it to forecast demand (c) evaluating approach in other datasets with frequent occurrences of zeros (d) building ensembles and use of AutoML techniques.

CRedit authorship contribution statement

Jože M. Rožanec: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Gašper Petelin:** Writing – review & editing, Writing – original draft, Validation. **João Costa:**

Validation, Software, Data curation. **Gregor Cerar:** Writing – review & editing, Writing – original draft, Software, Data curation. **Blaž Bertalančič:** Writing – original draft, Validation, Software. **Marko Guček:** Validation, Resources, Funding acquisition. **Gregor Papa:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Dunja Mladenčič:** Writing – review & editing, Validation, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

References

- Abanda, A., Mori, U., Lozano, J.A., 2019. A review on distance based time series classification. *Data Min. Knowl. Discov.* 33 (2), 378–412.
- Abraham, Z., Tan, P.-N., 2009. A semi-supervised framework for simultaneous classification and regression of zero-inflated time series data with application to precipitation prediction. In: 2009 IEEE International Conference on Data Mining Workshops. IEEE, pp. 644–649.
- Alzubaidi, L., Bai, J., Al-Sabaawi, A., Santamaría, J., Albahri, A.S., Al-dabbagh, B.S.N., Fadhil, M.A., Manoufali, M., Zhang, J., Al-Timemy, A.H., et al., 2023. A survey on deep learning tools dealing with data scarcity: definitions, challenges, solutions, tips, and applications. *J. Big Data* 10 (1), 46.
- Bagnall, A., Lines, J., Bostrom, A., Large, J., Keogh, E., 2017. The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Min. Knowl. Discov.* 31, 606–660.
- Balderas-Díaz, S., Guerrero-Contreras, G., Muñoz, A., Boubeta-Puig, J., 2024. Fusing temporal and contextual features for enhanced traffic volume prediction. In: *World Conference on Information Systems and Technologies*. Springer, pp. 74–84.
- Banerjee, N., Morton, A., Akartunali, K., 2020. Passenger demand forecasting in scheduled transportation. *European J. Oper. Res.* 286 (3), 797–810.
- Bertalančič, B., Fortuna, C., 2024. CARMEL: Capturing spatio-temporal correlations via time-series sub-window imaging for home appliance classification. *Eng. Appl. Artif. Intell.* 127, 107318.
- Bertalančič, B., Meža, M., Fortuna, C., 2022. Resource-aware time series imaging classification for wireless link layer anomalies. *IEEE Trans. Neural Networks Learn. Syst.* 1–13. <http://dx.doi.org/10.1109/TNNLS.2022.3149091>.
- Bradley, A.P., 1997. The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognit.* 30 (7), 1145–1159. [http://dx.doi.org/10.1016/S0031-3203\(96\)00142-2](http://dx.doi.org/10.1016/S0031-3203(96)00142-2), URL <http://www.sciencedirect.com/science/article/pii/S0031320396001422>.
- Cerar, G., Bertalančič, B., Fortuna, C., 2022. Resource-aware deep learning for wireless fingerprinting localization. *arXiv preprint arXiv:2211.01759*.
- Cohen, Jr., A.C., 1954. Estimation of the Poisson parameter from truncated samples and from censored samples. *J. Amer. Statist. Assoc.* 49 (265), 158–168.
- Cragg, J.G., 1971. Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica: J. Econ. Soc.* 829–844.
- Eckmann, J.-P., Kamphorst, S.O., Ruelle, D., et al., 1995. Recurrence plots of dynamical systems. *World Sci. Ser. Nonlinear Sci. Ser. A* 16, 441–446.
- Fang, X., Misra, S., Xue, G., Yang, D., 2012. Smart grid — The new and improved power grid: A survey. *IEEE Commun. Surv. Tutorials* 14 (4), 944–980. <http://dx.doi.org/10.1109/SURV.2011.101911.00087>.
- Foumani, N.M., Miller, L., Tan, C.W., Webb, G.I., Forestier, G., Salehi, M., 2023. Deep learning for time series classification and extrinsic regression: A current survey. *ACM Comput. Surv.*
- Franco, P., Martinez, J.M., Kim, Y.-C., Ahmed, M.A., 2021. IoT based approach for load monitoring and activity recognition in smart homes. *Ieee Access* 9, 45325–45339.
- Gopinath, R., Kumar, M., Joshua, C.P.C., Srinivas, K., 2020. Energy management using non-intrusive load monitoring techniques—state-of-the-art and future research directions. *Sustain. Cities Soc.* 62, 102411.
- Grinsztajn, L., Oyallon, E., Varoquaux, G., 2022. Why do tree-based models still outperform deep learning on typical tabular data? *Adv. Neural Inf. Process. Syst.* 35, 507–520.
- Gupta, A., Gupta, H.P., Biswas, B., Dutta, T., 2020. Approaches and applications of early classification of time series: A review. *IEEE Trans. Artif. Intell.* 1 (1), 47–61.
- Hemmerling, J., Pflugmacher, D., Hostert, P., 2021. Mapping temperate forest tree species using dense Sentinel-2 time series. *Remote Sens. Environ.* 267, 112743.
- Hyndman, R.J., et al., 2006. Another look at forecast-accuracy metrics for intermittent demand. *Foresight: Int. J. Appl. Forecast.* 4 (4), 43–46.
- Ibarra-Rojas, O.J., Delgado, F., Giesen, R., Muñoz, J.C., 2015. Planning, operation, and control of bus transport systems: A literature review. *Transp. Res. Part B: Methodol.* 77, 38–75.
- Ismail Fawaz, H., Forestier, G., Weber, J., Idoumghar, L., Muller, P.-A., 2019. Deep learning for time series classification: a review. *Data Min. Knowl. Discov.* 33 (4), 917–963.
- Iwana, B.K., Uchida, S., 2021. An empirical survey of data augmentation for time series classification with neural networks. *Plos One* 16 (7), e0254841.
- Kelly, J., Knottenbelt, W., 2015. The UK-DALE dataset, domestic appliance-level electricity demand and whole-house demand from five UK homes. *Sci. Data* 2 (1), 1–14.
- Klemenjak, C., Goldsborough, P., 2016. Non-intrusive load monitoring: A review and outlook. *arXiv preprint arXiv:1610.01191*.
- Lambert, D., 1992. Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics* 34 (1), 1–14.
- Li, Z.-C., Sheng, D., 2016. Forecasting passenger travel demand for air and high-speed rail integration service: A case study of Beijing-Guangzhou corridor, China. *Transp. Res. Part A: Policy Pr.* 94, 397–410.
- Matindife, L., Sun, Y., Wang, Z., 2021. Image-based mains signal disaggregation and load recognition. *Complex & Intell. Syst.* 7 (2), 901–927.
- McElfresh, D., Khandagale, S., Valverde, J., Prasad, C. V., Ramakrishnan, G., Goldblum, M., White, C., 2024. When do neural nets outperform boosted trees on tabular data? *Adv. Neural Inf. Process. Syst.* 36.
- MontazeriShatoori, M., Davidson, L., Kaur, G., Lashkari, A.H., 2020. Detection of doh tunnels using time-series classification of encrypted traffic. In: 2020 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress. DASC/PiCom/CBDCom/CyberSciTech, IEEE, pp. 63–70.
- Mullahy, J., 1986. Specification and testing of some modified count data models. *J. Econometrics* 33 (3), 341–365.
- Narayan, A., Reyes, F.A., Ren, M., Haoyong, Y., 2021. Real-time hierarchical classification of time series data for locomotion mode detection. *IEEE J. Biomed. Heal. Informatics* 26 (4), 1749–1760.
- Petelin, G., Hribar, R., Papa, G., 2023. Models for forecasting the traffic flow within the city of Ljubljana. *Eur. Transp. Res. Rev.* 15 (1), 30.
- Pirnat, A., Bertalančič, B., Cerar, G., Mohorič, M., Fortuna, C., 2024. Energy efficient deep multi-label ON/OFF classification of low frequency metered home appliances. *IEEE Access* 12, 51966–51981. <http://dx.doi.org/10.1109/ACCESS.2024.3382830>.
- Ridi, A., Gisler, C., Hennebert, J., 2014. A survey on intrusive load monitoring for appliance recognition. In: 2014 22nd International Conference on Pattern Recognition. IEEE, pp. 3702–3707.
- Rodrigues, P., Martins, A., Kalakou, S., Moura, F., 2020. Spatiotemporal variation of taxi demand. *Transp. Res. Procedia* 47, 664–671.
- Rožanec, J.M., Fortuna, B., Mladenčič, D., 2022. Reframing demand forecasting: a two-fold approach for lumpy and intermittent demand. *Sustainability* 14 (15), 9295.
- Rožanec, J.M., Šircelj, B., Onat, M.K., Sarantinoudis, N., Arampatzis, G., Fortuna, B., Mladenčič, D., 2023. Machine-learning-based soft sensors for LPG distillation columns quality monitoring: Comparing strategies for zero-inflated data. In: 2023 International Conference on Electrical, Computer and Energy Technologies. ICECET, IEEE, pp. 1–6.
- Simonyan, K., Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.
- Suh, S., Rey, V.F., Bian, S., Huang, Y.-C., Rožanec, J.M., Ghinani, H.T., Zhou, B., Lukowicz, P., 2023. Worker activity recognition in manufacturing line using near-body electric field. *IEEE Internet Things J.*
- Susto, G.A., Cenedese, A., Terzi, M., 2018. Time-series classification methods: Review and applications to power systems data. *Big Data Appl. Power Syst.* 179–220.
- Tobin, J., 1958. Estimation of relationships for limited dependent variables. *Econometrica: J. Econ. Soc.* 24–36.
- Truong, T.M.T., Ngoc, A.M., 2020. The city-airport connection, implications for urban transport planning in motorcycle-dominated cities. *Transp. Res. Procedia* 48, 3232–3244.

- Vansteenkoven, P., Melis, L., Aktaş, D., Montenegro, B.D.G., Vieira, F.S., Sørensen, K., 2022. A survey on demand-responsive public bus systems. *Transp. Res. Part C: Emerg. Technol.* 137, 103573.
- Youden, W.J., 1950. Index for rating diagnostic tests. *Cancer* 3 (1), 32–35.
- Zhou, Y., Deng, H., Zeng, Y., Chen, H., Lian, Z., 2019. Holiday travel pattern forecast based on machine learning algorithm. In: 2019 IEEE 4th Advanced Information Technology, Electronic and Automation Control Conference. IAEAC, IEEE, pp. 2288–2291.
- Zoha, A., Gluhak, A., Imran, M.A., Rajasegarar, S., 2012. Non-intrusive load monitoring approaches for disaggregated energy sensing: A survey. *Sensors* 12 (12), 16838–16866.